

Surrogate Modeling of a Binary Distillation Column

Using DWSIM and Machine Learning

System: Benzene - Toluene

Thermodynamic Model: Peng-Robinson (PR)

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1. Problem Statement

The objective of this work is to develop a surrogate model for a binary distillation column separating Benzene and Toluene. Rigorous process simulations in DWSIM are accurate but computationally expensive. A machine learning surrogate model replaces these simulations with a fast, accurate predictor trained on data generated from DWSIM.

The surrogate model predicts the following four output variables:

- Distillate purity ($x_{D_benzene}$): mole fraction of Benzene in the distillate stream
- Bottoms purity ($x_{B_toluene}$): mole fraction of Toluene in the bottoms stream
- Condenser duty (Q_{C_kW}): heat removed in the condenser, in kW
- Reboiler duty (Q_{R_kW}): heat supplied in the reboiler, in kW

The following seven operating conditions are used as inputs:

- Feed temperature (T_{feed_K}) in K
- Feed pressure (P_{feed_Pa}) in Pa
- Feed composition ($x_{F_benzene}$): mole fraction of Benzene in feed
- Number of equilibrium stages (N_{stages})
- Feed stage location ($feed_stage$)
- Reflux ratio ($reflux_ratio$)
- Bottoms molar flow rate ($bottoms_flow_mols$) in mol/s

2. DWSIM Flowsheet and Model Description

A rigorous distillation column was configured in DWSIM using the built-in Distillation Column unit operation. The column separates a binary mixture of Benzene (light component) and Toluene (heavy component) at atmospheric pressure using the Peng-Robinson equation of state.

Flowsheet configuration:

- Thermodynamic package: Peng-Robinson (PR) equation of state
- Feed molar flow rate: 100 mol/s (fixed for all simulations)
- Operating pressure: 101,325 Pa (1 atm, fixed for all simulations)
- Condenser type: Total condenser
- Condenser specification: Reflux ratio
- Reboiler specification: Bottoms molar flow rate

The flowsheet contains four named objects: Feed (material stream), Distillation Column (unit operation), Distillate (product stream), and Bottoms (product stream). The column was automated using the DWSIM Automation API (Automation3 class) via Python.NET. For each simulation, parameters were set on the live flowsheet using the Specs dictionary (key C for reflux ratio with SType Stream_Ratio, key R for bottoms flow rate with SType Product_Molar_Flow_Rate) before calling CalculateFlowsheet2 to solve.

3. Input and Output Variables

3.1 Input Variables

Variable	Symbol	Unit	Description
Feed Temperature	T_feed_K	K	Temperature of the feed stream
Feed Pressure	P_feed_Pa	Pa	Pressure of feed (fixed at 101,325 Pa)
Feed Composition	xF_benzene	mol fraction	Mole fraction of Benzene in feed
Number of Stages	N_stages	—	Total equilibrium stages in the column
Feed Stage	feed_stage	—	Stage number where feed enters
Reflux Ratio	reflux_ratio	—	Ratio of liquid returned to distillate taken off
Bottoms Flow Rate	bottoms_flow_mols	mol/s	Molar flow rate of the bottoms product

3.2 Output Variables

Variable	Symbol	Unit	Description
Distillate Purity	xD_benzene	mol fraction	Mole fraction of Benzene in distillate
Bottoms Purity	xB_toluene	mol fraction	Mole fraction of Toluene in bottoms
Condenser Duty	QC_kW	kW	Heat duty of the condenser
Reboiler Duty	QR_kW	kW	Heat duty of the reboiler

4. Dataset Generation Strategy

The dataset was generated by automating DWSIM simulations through the DWSIM Automation API. A Python script set input parameters on the live flowsheet, solved the column for each parameter set, and recorded the output values from the solved streams.

4.1 Sampling Method

Latin Hypercube Sampling (LHS) was used to generate the parameter sets. LHS divides each input dimension into equal intervals and places exactly one sample per interval, ensuring uniform coverage of the entire input space. For a 6-dimensional input space, LHS with 3000 samples provides better coverage than a brute-force grid search while using far fewer simulations.

4.2 Dataset Summary

Parameter	Value
Sampling method	Latin Hypercube Sampling (LHS)
Total simulations attempted	3000
Successful simulations	2923
Failed simulations (convergence failures)	77
Final dataset size	2923 rows x 11 columns
Random seed	42

5. Range of Operating Conditions

Variable	Minimum	Maximum	Unit
T_feed_K	340	385	K
xF_benzene	0.20	0.80	mol fraction
N_stages	10	20	—
feed_stage	2	N - 1	—
reflux_ratio	1.5	4.0	—
bottoms_flow_mols	30	70	mol/s
P_feed_Pa	101,325	101,325	Pa (fixed)

The feed stage location was computed as a fraction between 25% and 65% of the total number of stages, ensuring the feed is always within the column and never at the condenser or reboiler stage.

6. Data Preprocessing

The following preprocessing steps were applied to the raw simulation data:

- Removal of failed rows: 77 simulations that did not converge were excluded, leaving 2923 valid data points.
- Physical validity check: rows where $x_{D_benzene}$ or $x_{B_toluene}$ fell outside the range (0.01, 0.9999) were rejected.
- Correlation verification: all input variables were confirmed to have non-zero correlation with at least one output, confirming all parameters were correctly written to the DWSIM flowsheet before each solve.
- Train-test split: 80% training (2338 samples) and 20% test (585 samples), using random seed 42.
- Feature scaling: StandardScaler (zero mean, unit variance) was applied for Linear Regression, Polynomial Regression, Neural Network, and SVM. Tree-based models (Random Forest, XGBoost, Gradient Boosting) did not require scaling.
- No missing values were present in the dataset.

7. Machine Learning Methods Implemented

Seven machine learning models were trained and compared.

MultiOutputRegressor from scikit-learn was used to extend all single-output regressors to predict all four output variables simultaneously.

Model	Key Hyperparameters	Scaling
Linear Regression	Default settings	Yes
Polynomial Regression (d=2)	degree=2, interaction terms included	Yes
Random Forest	n_estimators=200, min_samples_leaf=2, n_jobs=-1	No
XGBoost	n_estimators=300, max_depth=6, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8	No
Neural Network (MLP)	hidden_layers=(128,128,64), ReLU, early_stopping=True, max_iter=1000	Yes
Gradient Boosting	n_estimators=200, max_depth=5, learning_rate=0.05, subsample=0.8	No
SVM (RBF Kernel)	C=10, gamma=scale, epsilon=0.01	Yes

8. Model Training and Validation Procedure

- All models were trained on 2338 samples (80%) and evaluated on a held-out test set of 585 samples (20%).
- The test set was not used during training or hyperparameter selection.
- Neural Network used an additional internal validation split (10% of training data) with early stopping to prevent overfitting.
- Three metrics were computed for each model on each output variable: R2 Score (proportion of variance explained), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).
- A learning curve analysis was performed on Gradient Boosting using 5-fold cross-validation to assess overfitting.

9. Performance Comparison of Models

9.1 Overall Comparison

Rank	Model	Mean R2	Mean RMSE	Grade
1	Gradient Boosting	0.9955	0.0808	Excellent
2	Neural Network (MLP)	0.9937	0.0408	Excellent
3	Random Forest	0.9903	0.1248	Excellent
4	SVM (RBF)	0.9903	0.1144	Excellent
5	XGBoost	0.9876	0.1216	Good
6	Polynomial Reg (d=2)	0.9670	0.1850	Good
7	Linear Regression	0.8765	0.3107	Fair

9.2 Per-Output Performance of All Models

Model	xD R2	xB R2	QC R2	QR R2
Linear Regression	0.8286	0.8085	0.9743	0.8948
Polynomial Reg (d=2)	0.9681	0.9662	0.9993	0.9346
Random Forest	0.9929	0.9940	0.9962	0.9783
XGBoost	0.9851	0.9868	0.9925	0.9861
Neural Network	0.9895	0.9870	0.9994	0.9990
Gradient Boosting	0.9962	0.9967	0.9988	0.9904
SVM (RBF)	0.9943	0.9943	0.9998	0.9729

9.3 Detailed Metrics for Best Model (Gradient Boosting)

Output Variable	R2 Score	RMSE	MAE
xD_benzene	0.9962	0.0127	0.0074
xB_toluene	0.9967	0.0116	0.0076
QC_kW	0.9988	0.0611	0.0480
QR_kW	0.9904	0.2380	0.1611

10. Physical Consistency and Robustness

10.1 Physical Consistency

The surrogate model was tested by varying one input at a time while fixing all others at base values: $T=365$ K, $x_F=0.5$, $N=15$, $\text{feed_stage}=7$, $RR=2.5$, $B=50$ mol/s.

Test	Predicted Values	Expected Trend	Result
$x_F: 0.2 \rightarrow 0.5 \rightarrow 0.8$	$x_D: 0.4010 \rightarrow 0.9584 \rightarrow 0.9983$	x_D increases with x_F	Correct
$RR: 1.5 \rightarrow 2.5 \rightarrow 4.0$	$x_D: 0.9122 \rightarrow 0.9584 \rightarrow 0.9683$	x_D increases with RR	Correct
$B: 30 \rightarrow 50 \rightarrow 70$ mol/s	$QR: 7.2728 \rightarrow 4.7870 \rightarrow 3.1598$	QR decreases as more bottoms taken	Physically consistent
$N: 10 \rightarrow 15 \rightarrow 20$	$x_D: 0.9247 \rightarrow 0.9584 \rightarrow 0.9642$	x_D increases with N	Correct

Note on bottoms flow test: as bottoms flow increases from 30 to 70 mol/s (with distillate decreasing correspondingly), the reboiler duty decreases. This is physically consistent because a higher bottoms flow rate means less stringent separation is required at the bottom of the column, reducing the reboiler energy requirement.

10.2 Feature Importance

Feature importance was extracted from the Gradient Boosting model by averaging importances across all four output estimators:

Rank	Feature	Importance	Physical Interpretation
1	xF_benzene	0.4084	Feed composition is the primary driver of separation performance
2	bottoms_flow_mols	0.3421	Bottoms flow controls the split ratio and energy balance
3	reflux_ratio	0.1502	Reflux ratio controls liquid-vapor contact and purity
4	T_feed_K	0.0980	Feed thermal state affects the column energy balance
5	N_stages	0.0012	Lower importance within the tested range of 10-20 stages
6	feed_stage	0.0001	Feed location has the smallest effect within the tested range

10.3 Overfitting and Generalization

Metric	Value
Training R2 (Gradient Boosting)	0.9993
Cross-validation R2 (5-fold)	0.9956
Gap (Train - Validation)	0.0037
Verdict	No overfitting detected

The gap of 0.0037 between training and validation R2 is negligible, confirming the model generalizes well to unseen data. The learning curve shows both training and validation scores converging at high R2 values as training size increases.

11. Final Conclusion

Gradient Boosting is selected as the best surrogate model based on the following:

- Highest prediction accuracy among all seven models: Mean $R^2 = 0.9955$ across all four output variables.
- No overfitting: training and cross-validation R^2 gap of only 0.0037.
- Physical consistency: predicted trends are correct for all four tested input variations.
- Strong per-output performance: R^2 above 0.99 for x_D , x_B , and Q_C ; $R^2 = 0.9904$ for Q_R .
- Stable and smooth predictions across the full operating range.

Among all models, tree-based ensemble methods (Gradient Boosting, Random Forest) outperformed linear methods, confirming that the relationships between distillation operating conditions and performance variables are nonlinear. Linear Regression serves only as a baseline and is insufficient for accurate surrogate modeling of distillation systems. Neural Network and SVM also performed well but were outperformed by Gradient Boosting on the overall mean R^2 metric.

The surrogate model reduces prediction time from seconds (DWSIM rigorous solve) to microseconds (model inference), while maintaining over 99% accuracy on unseen data. This makes it suitable for applications such as real-time process monitoring, rapid design space exploration, and optimization.